

Pranav Prashant Shewale

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EDUCATION

Vellore Institute of Technology, Chennai Campus, BTech in CSE – GPA: 8.27/10.0	Jul 2024 – Present
SBS College, Karad, PCM (HSC Board) – Percentage: 84.66%	2022 – 2024
Tilak Highschool, Karad, SSC Board – Percentage: 97.40%	2020 – 2022

WORK EXPERIENCE

Bonzai AI Engineer / SDE Intern – Remote	May 2026 – Present
- Built an automated Python data pipeline to extract and structure Shopify e-commerce data into an AWS database, providing the clean datasets required for AI model training. - Designed the AWS backend infrastructure to support AI workloads, building scalable APIs for model inference and setting up automated CI/CD pipelines for reliable deployments.	
Sentrix Robotics Pvt. Ltd. CO-Founder Researcher – Chennai, India	Mar 2026 – Present
Developing data pipelines and perception systems for physical AI using Python and PyTorch, enabling structured sensory data processing to address robotic tactile blindness.	
UrbanDienst Computer Vision Intern – Remote	Nov 2025 – Feb 2026
Deployed YOLOv8 on MemryX edge accelerators with quantization, achieving 45 FPS, 22 ms latency, and sub-50 MB memory for real-time inference.	
IIT Mandi Artificial Intelligence + Robotics Intern – Mandi, Himachal Pradesh	Apr 2025 – Oct 2025
Co-authored “TongueSight”: built AI diagnostic pipeline using YOLOv8 and Mask2Former with EfficientNet-B0/ResNet feature extraction to quantify tongue biomarkers; achieved 92.0% accuracy, outperforming TongueNet and Cv-Swin Transformer.	

PUBLICATIONS

- TongueSight: AI-Powered Tongue Analysis for Digitizing Traditional Medicine Diagnostics** | Published at the 14th International Conference on Graphics, Vision and Image Processing (ICVGIP 2025) – [Paper](#)
 - Co-authored hybrid deep learning pipeline (YOLOv8 + Mask2Former/SegNet/EfficientNet-B0) achieving **92.0%** accuracy with modular design for **mobile/edge deployment**.

PROJECTS

-  **VLM-Prompted Segmentation for Drywall Defect Detection**
 - Built prompt-driven segmentation pipeline using SAM/Grounding DINO for defect localization; enabled zero-shot generalization for real-world inspection workflows.
-  **ColdDraft: AI Research Discovery & Outreach Engine**
 - Engineered NLP-based matching (TF-IDF, cosine similarity) with multi-threaded scraping and Llama 3 agent pipelines; automated personalized outreach via FastAPI.
-  **Multi-Agent Task Orchestration System**
 - Developed LangGraph/LangChain multi-agent system with autonomous task routing, LLM-based refinement loops, and real-time Streamlit monitoring.

TECHNICAL SKILLS

- Programming Languages:** Python, C/C++, Java, SQL
- ML/DL & Computer Vision:** PyTorch, TensorFlow, YOLOv8, ViT, DETR, Mask2Former, StrongSORT, OpenCV, ONNX, MobileNetV2, CNN, dlib
- Robotics & Autonomy:** Reinforcement Learning (PPO), Multi-Agent Systems, Edge AI, Sensor Processing
- Edge Deployment:** Raspberry Pi, MemryX, Docker, FastAPI, WebSockets, TensorFlow Lite